Skewness, Moments & Kurtosis

Skewness :-

consider the following two distributions

cont class	freq.
0 - 5	10
5 - 10	30
10 - 15	60
15 - 20	60
20 - 25	30
25 - 30	10

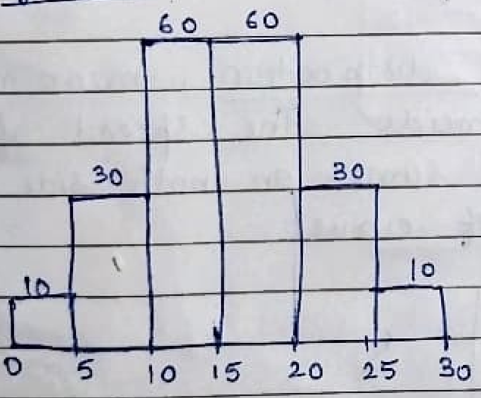
cont class	freq
0 - 5	10
5 - 10	40
10 - 15	30
15 - 20	90
20 - 25	20
25 - 30	10.

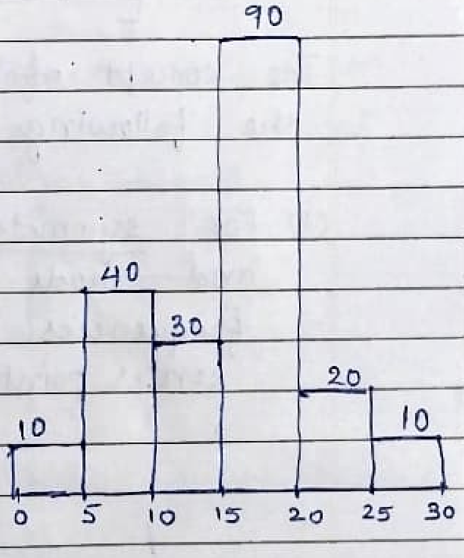
both the distributions has same mean and S.D

$\overline{X} = 15$

$\sigma = 6$

Graphs





Both the distributions have same mean and S.D but distribution are not alike in nature. Left distribution is symmetrical and right is assymetrical. Skewness help us to distinguish different type of distribution, with the help of symmetry.

* when series is not symmetrical it is said to be assymetrical or skewed.

* Skewness is concerned with the shape of the curve not size.

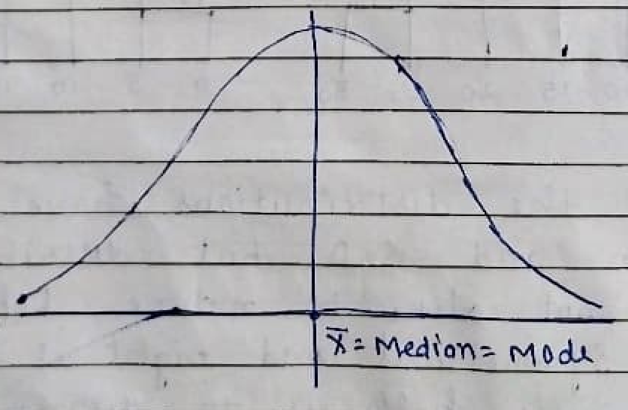
* Measure of skewness tell us the direction and extent of skewness. In symmetrical distribution then mean, median and mode are identical. The more the mean, moves away from the mode, the assymetry or skewness is more.

For Moderately skewed distribution, the following relation hold among the three measures of central tendency.

[Mean − Mode = 3(Mean − Median)]

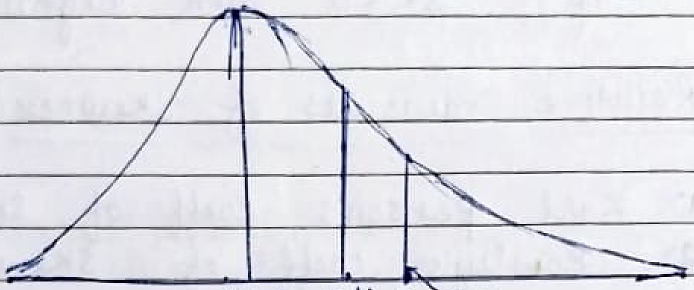
The concept of skewness will be clear from the following diagrams:-

(1) For symmetrical distribution, mean, median and mode coincide. The spread of frequencies is same on both side of center point of curve.



(4) Positively skewed distribution or Skewed to right

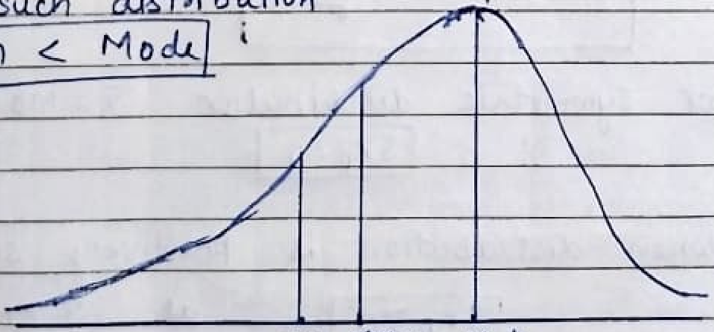
If the frequency curve of a distribution has longer tail towards the right of the central maximum than to the left, the distribution is said to be skewed to the right or said to have positive Skewness. In such distribution $\boxed{\text{Mean} > \text{Mode}}$



Mode Median Mean $\bar{x}$

(5) Negatively skewed or Skewed to left :-

If it has longer tail towards the left of central maximum than to the right. It is said to negatively skewed. In such distribution $\boxed{\text{Mean} < \text{Mode}}$



Mean $\bar{x}$ Median Mode

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Measures of Skewness :-

(1) Absolute measure of Skewness (SK) :-

$$SK = \bar{X} - \text{Mode}$$

if SK = 0, symmetrical distri
if SK > 0 then, positively skewed
if SK < 0 then negatively skewed

(2) Relative measures of skewness :-

(a) Karl pearson's coeff. of skewness
(b) Bowley's coeff. of skewness
(c) Measure of skewness based on moments

(a) Karl Pearson's coeff. of skewness :-

$$SK_p = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}}$$

$$\boxed{SK_p = \frac{\bar{X} - Mo}{\sigma}}$$

* for symmetric distribution $\bar{X} = Mo$ so

$\boxed{SK_p = 0}$

* when distribution is positively skewed

$\boxed{SK_p > 0}$ as $\bar{X} > \text{Mode}$

* for negatively skew

$\boxed{SK_p < 0}$ as $\bar{X} < \text{Mode}$

The degree of skewness can be measured by numerical value say 0.8 or .2 etc.

using Mode = 3 Median − 2 Mean

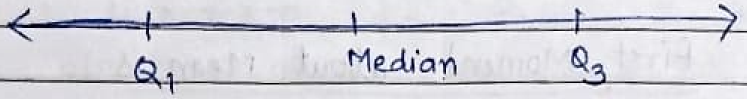
$$SK_p = \frac{3(\bar{x} - \text{Median})}{\sigma}$$

The value of SK_p varies b/w ± 3, however in practice it is rare that the value of SK_p exceeds ± 1.

Note:- This method of measuring skewness can't be used when mode is illdefined.

(b) Bowley's coeff. of skewness:-

It is Based of Quartiles.



In symmetrical distribution, first and third Quartiles are equidistant from median

ie $Q_3 - \text{Median} = \text{Median} - Q_1$] ie $\boxed{SK_B = 0}$ for sym. dis.

ie $Q_3 + Q_1 - 2\,\text{Median} = 0$

For skewed distribution:-

$$\boxed{SK_B = \frac{Q_3 + Q_1 - 2\,\text{Median}}{Q_3 - Q_1}}$$

$$\text{or } SK_B = \frac{(Q_3 - \text{Median}) - (\text{Median} - Q_1)}{(Q_3 - \text{Median}) + (\text{Median} - Q_1)}$$

For positively skewed distribution, Q_3 will be farther from median than Q_1 is from median

so $\boxed{SK_B > 0}$.

Slly for negatively skewed $\boxed{SK_B < 0}$

This is also called quartile measure of skewness and value of the coefficient varies b/w ± 1.

(5) Measure of Skewness based on Moments :-

The r^{th} moment about the mean is define as

$$M_r = \frac{\Sigma (x-\bar{x})^r}{N}, \quad r = 0, 1, 2.$$

For continous grouped data it is given by

$$M_r = \frac{\Sigma f (x-\bar{x})^r}{N}, \quad r = 0, 1, 2, --$$

ie

First Moment about Mean is

$$\mu_1 = \frac{\Sigma (x-\bar{x})}{N}$$

(As sum of deviation from arithmatic Mean is always 0 ie $\mu_1 = 0$)

Second Moment

$$\mu_2 = \frac{\Sigma (x-\bar{x})^2}{N} = \sigma^2 \text{ variance}$$

Third Moment

$$\mu_3 = \frac{\Sigma (x-\bar{x})^3}{N}$$

Fourth Moment

$$\mu_4 = \frac{\Sigma (x-\bar{x})^4}{N}$$

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Sly for frequency distribution

$\mu_1 = \frac{\Sigma f(x-\bar{x})}{N}$, $\mu_2 = \frac{\Sigma f(x-\bar{x})^2}{N} = \sigma^2$

$\mu_3 = \frac{\Sigma f(x-\bar{x})^3}{N}$ $\mu_4 = \frac{\Sigma f(x-\bar{x})^4}{N}$

Moment in statistics are used to define various characterstics of frequency distribution like central tendency, variation, skewness and kurtosis.

Arithmatic mean is actually first moment about origin ie $\mu_1 = \frac{\Sigma x - 0}{N} = \frac{\Sigma x}{N} = $ Mean, the second moment about mean ie $\mu_2 = \frac{\Sigma(x-\bar{x})^2}{N}$ is variance. etc.

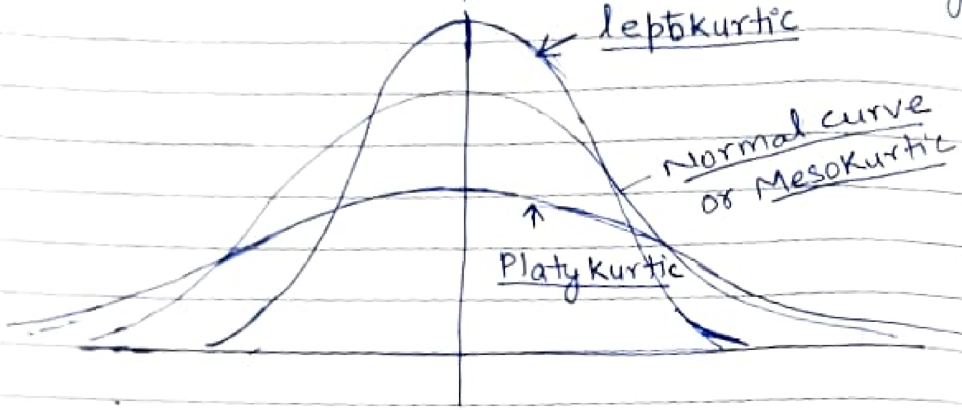
There are two important constant of distribution are calculated from μ_2, μ_3 & μ_4.

$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$ called coefficient of skewness (also called measure of skewness based on moments).

$\beta_2 = \frac{\mu_4}{\mu_2^2}$, called coeff. of Kurtosis.

Kurtosis :- Kurtosis refer to the degree of flatness or Peakedness of a frequency curve about the mode.
In other words measure of Kuratosis tells the extent to which a distribution is more peaked or flat-topped than the normal curve.

The Normal curve, not very high peaked, is itself known as 'mesokurtic'.
If a curve is more peaked than normal curve it is known as 'leptokurtic'.
If a curve is more flat topped than the normal curve, it is called platykurtic



Kurtosis is measured by β_2

$$\text{coeff. } \beta_2 = \frac{\mu_4}{\mu_2^2} \quad \text{or} \quad \gamma_2 = \beta_2 - 3$$

(i) If $\beta_2 = 3$ ie $\gamma_2 = 0$
For a Normal curve value of $\beta_2 = 3$ ie $\gamma_2 = 0$

ie If $\beta_2 = 3$ or $\gamma_2 = 0$, curve is <u>Mesokurtic</u>.

(ii) If $\beta_2 > 3$, ie $\gamma_2 > 0$, then curve is more peaked than normal curve ie called <u>leptokurtic</u>.

(iii) If $\beta_2 < 3$ ie $\gamma_2 < 0$ then curve is less peaked than normal curve ie called <u>platykurtic</u>.

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Moments about arbitrary origin :-

when the actual mean is in fractions it is difficult to calculate moment about mean, so we ~~to~~ can calculate moment about an arbitrary origin A then convert these moment into moment about actual mean.

Moment about arb origin A are

$$\mu_1' = \frac{\Sigma(x-A)}{N}, \quad \mu_2' = \frac{\Sigma(x-A)^2}{N}, \quad \mu_3' = \frac{\Sigma(x-A)^3}{N}$$

$$\mu_4' = \frac{\Sigma(x-A)^4}{N}$$

For freq. distribution

$$\mu_1' = \frac{\Sigma f(x-A)}{N}, \quad \mu_2' = \frac{\Sigma f(x-A)^2}{N} \text{ or } \frac{\Sigma f d^2}{N}$$

or

$$\mu_1' = \frac{\Sigma f d}{N}, \; d = x-A \quad \text{or} \quad \frac{\Sigma f d'^2}{N} \times h, \; d' = \frac{x-A}{h}$$

or $\mu_1' = \frac{\Sigma f d'}{N} \times h$ where $d' = \frac{x-A}{h}$

Slly ~~other~~ 3rd & 4th moment can be written

Now Moments about Mean can be calculated from moments about arb. origin :-

$$\mu_1 = \mu_1' - \mu_1' = 0$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3$$

$$\mu_4 = \mu_4' - 4\mu_1'\mu_3' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$$

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Q-1 Calculate Pearson's coeff. of skewness from the following.

X	12.5	17.5	22.5	27.5	32.5	37.5	42.5	47.5
f	28	42	54	108	129	61	45	33

Ans $SK_p = \frac{\text{Mean} - \text{Mode}}{\sigma}$.

X	f	$d = X - 32.5/5$	fd	fd^2
12.5	28	-4	-112	448
17.5	42	-3	-126	378
22.5	54	-2	-108	216
27.5	108	-1	-108	108
32.5	129	0	0	0
37.5	61	1	61	61
42.5	45	2	90	180
47.5	33	3	99	297.

Mean $\bar{X} = A + \frac{\Sigma fd}{N} \times h$ $h = 5$

$\bar{x} = 30.46$

$S.D = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times h = 8.96$

Mode = 32.5

$SK_p = \frac{30.46 - 32.5}{8.96} = -.228$